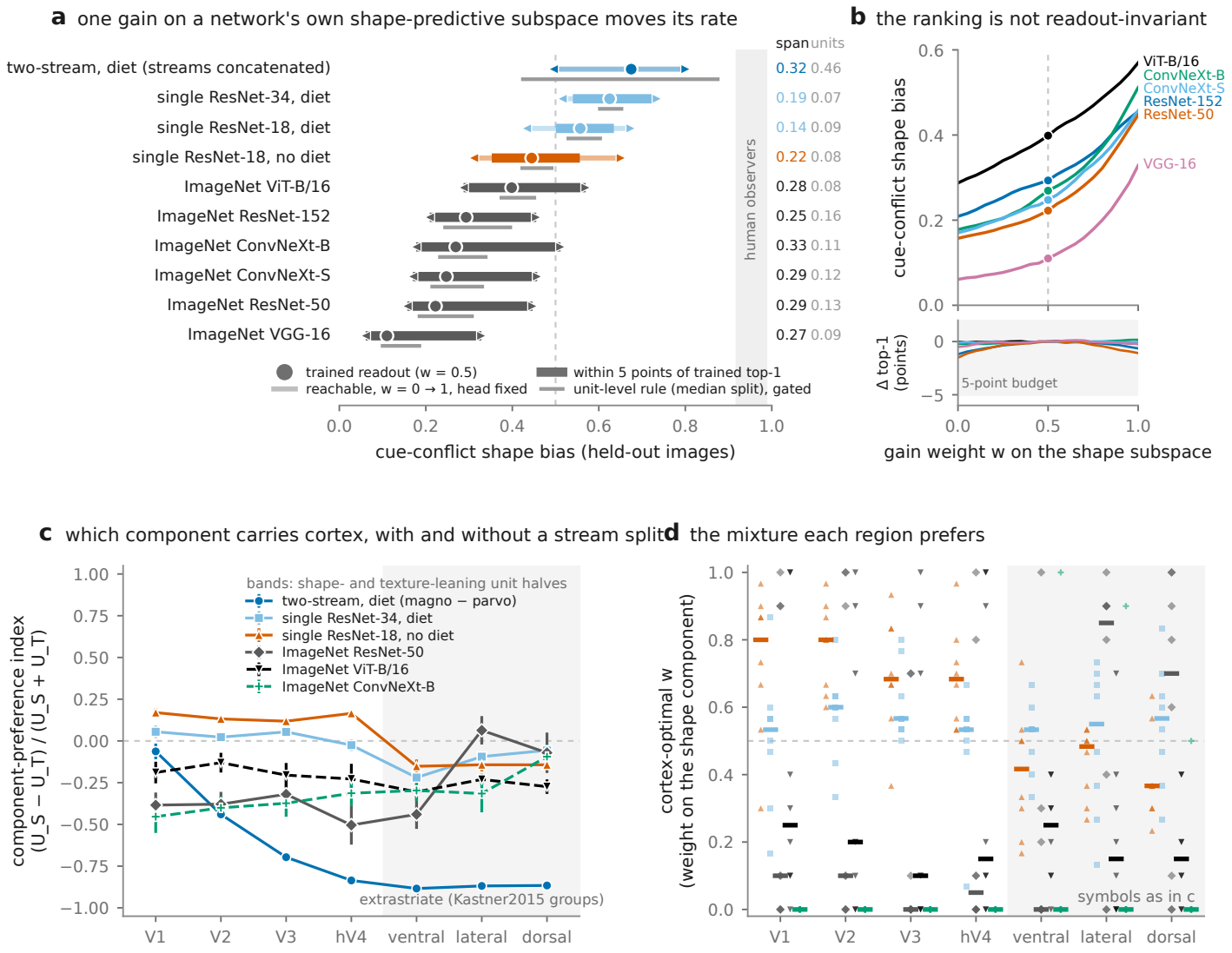
a one gain on a network's own shape-predictive subspace moves its rate
two-stream, diet (streams concatenated)
single ResNet-34, diet
single ResNet-18, diet
single ResNet-18, no diet
ImageNet ViT-B/16
ImageNet ResNet-152
ImageNet ConvNeXt-B
ImageNet ConvNeXt-S
ImageNet ResNet-50
ImageNet VGG-16
span units
0.32 0.46
0.19 0.07
0.14 0.09
0.22 0.08
0.28 0.08
0.25 0.16
0.33 0.11
0.29 0.12
0.29 0.13
0.27 0.09
human observers
trained readout (w = 0.5)
within 5 points of trained top-1
reachable, w = 0 → 1, head fixed
unit-level rule (median split), gated
0.0 0.2 0.4 0.6 0.8 1.0
cue-conflict shape bias (held-out images)
b the ranking is not readout-invariant
cue-conflict shape bias
ViT-B/16
ConvNeXt-B
ConvNeXt-S
ResNet-152
ResNet-50
VGG-16
Δ top-1 (points)
5-point budget
gain weight w on the shape subspace
c which component carries cortex, with and without a stream split
component-preference index (U_S − U_T) / (U_S + U_T)
bands: shape- and texture-leaning unit halves
two-stream, diet (magno − parvo)
single ResNet-34, diet
single ResNet-18, no diet
ImageNet ResNet-50
ImageNet ViT-B/16
ImageNet ConvNeXt-B
extrastriate (Kastner2015 groups)
V1 V2 V3 hV4 ventral lateral dorsal
d the mixture each region prefers
cortex-optimal w (weight on the shape component)
symbols as in c
V1 V2 V3 hV4 ventral lateral dorsal